



Geochronology, petrology and Hf–S isotope geochemistry of the newly-discovered Xiarihamu magmatic Ni–Cu sulfide deposit in the Qinghai–Tibet plateau, western China

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ABSTRACT

This paper reports the first set of data for the newly-discovered Xiarihamu magmatic Ni–Cu sulfide deposit in the Eastern Kunlun Paleozoic arc terrane which is located in the northern part of the Qinghai–Tibet plateau. An on-going drilling campaign reveals ~100 million tons of sulfide mineralization with the average grade of 0.8 wt.% Ni and 0.1 wt.% Cu for the deposit. This makes the Xiarihamu deposit one of the 20 largest magmatic Ni–Cu sulfide deposits in the world and the largest ever found in arc settings. The deposit is hosted in a small ultramafic body intruding older gabbroic and metamorphic rocks. New zircon U–Pb isotope age data reveal that the ultramafic body (411.6 ± 2.4 Ma) is ~20 Ma younger than the host gabbroic intrusion (431.3 ± 2.1 Ma). The ultramafic body is composed predominantly of lherzolite and olivine websterite, with minor dunite, websterite and orthopyroxenite. Mineralization mainly occurs as sub-horizontal to gently dipping ($<30^\circ$) disseminated sulfide zones that are generally concordant with the lithological structure of the ultramafic intrusion. The lateral extension and thickness of individual mineralized zones are up to ~200 m and ~100 m, respectively. Sulfide mineral assemblages are composed of pyrrhotite, pentlandite and chalcopyrite. The Xiarihamu ultramafic rocks show light REE enrichments and pronounced negative Nb anomalies, plus significant Ca-depletion in olivine (<700 ppm Ca), which are characteristic of many arc basalts in the world. Olivine crystals in the Xiarihamu ultramafic rocks have relatively primitive compositions, with Fo contents up to 90 mol%, close to the mantle value. The contrasting Ni contents of olivine crystals with similar Fo contents from different sulfide-mineralized zones in a single drill core indicate that at least two pulses of sulfide-laden magma with different Ni compositions were involved in the development of the deposit. Estimated parental magma for the Xiarihamu lherzolites contains 52.4 wt.% SiO_2 and 9.8 wt.% MgO, which are within the ranges of boninites worldwide, supporting the interpretation that the Xiarihamu sulfide ore-bearing ultramafic intrusion is the product of subduction-related boninitic magmatism. The $\epsilon_{\text{Hf}}(t)$ values of zircon crystals from the Xiarihamu ultramafic rocks vary from 1 to 5, indicating minor crustal contamination of the parent magma. The involvement of external sulfur in the genesis of the Xiarihamu magmatic sulfide deposit is evident from higher $\delta^{34}\text{S}$ values of sulfide ores (3.5 to 6.8‰) than typical mantle value ($0 \pm 2\text{‰}$), although the actual contaminants are yet to be identified.

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1. Introduction

The Xiarihamu magmatic sulfide deposit, which occurs in the northern part of the Tibet–Qinghai plateau, was discovered in 2011 by a state-funded exploration program to test a gossan (S.-J. Li *et al.*, 2012). An on-going drilling campaign has delineated a total resource of ~100 million tons of ore with an average grade of 0.8 wt.% Ni and

0.1 wt.% Cu for the deposit (personal communication with an exploration geologist in charge, June 2014). This makes the Xiarihamu deposit one of the 20 largest magmatic Ni deposits in the world. The Xiarihamu discovery thus represents the most significant discovery worldwide since the discovery of the Voisey's Bay Ni–Cu–Co deposit in Labrador, Canada two decades ago. Among the 20 known largest magmatic sulfide deposits in the world, the Xiarihamu deposit is the only one that appears to have formed in an active convergent plate margin, all others clearly formed in stable intraplate tectonic settings except the meteorite impact melt-related deposits in Sudbury, Ontario, Canada (see a summary in

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Naldrett, 2011). The Xiarihamu deposit thus provides a unique opportunity to study the formation of magmatic sulfide deposits in such an unusual tectonic setting for magmatic sulfide deposits. In this communication we report the first set of petrological and geochemical data for the deposit, including a zircon U–Pb isotope age, Hf isotopes, olivine chemistry, whole-rock compositions, and S isotopes. We use these data to construct a conceptual ore-genesis model which bears some implications for mineral exploration at both regional and global scales.

2. Geological setting

The Xiarihamu Ni–Cu sulfide deposit is hosted in a mafic–ultramafic complex in the East Kunlun orogenic belt in the northern part of the Qinghai–Tibet plateau (Fig. 1). Several magmatic Ni–Cu sulfide occurrences with different ages have been found north of the East Kunlun orogenic belt (Fig. 2a). These include the 367 Ma Niubiziling occurrence in the western part of the Qaidam block (Ling et al., 2014), the 441–443 Ma Yaqi and Yulonggou occurrences in the southeastern part of the Qilian block (Z.-W. Zhang et al., 2014), and the 831 Ma Jinchuan deposit in the southern part of the Alxa block (Zhang et al., 2010). The on-going drilling results show that the Niubiziling, Yaqi and Yulonggou Ni–Cu sulfide occurrences are all too small to be economically valuable under current market conditions. In contrast, the Jinchuan deposit, which has been mined since the early 70s, is one of the three largest magmatic Ni deposits/camps in the world (Li and Ripley, 2011).

The East Kunlun orogenic belt is the eastern segment of the E–W trending Kunlun Paleozoic–Mesozoic orogenic belt which extends for >2000 km from central China to eastern Pakistan. The western segment of the Kunlun orogenic belt is referred to as the West Kunlun orogenic belt in the literature (e.g., Xiao et al., 2005). These two segments have been displaced from each other by ~475 km (Fig. 2a) since the middle Oligocene by the active sinistral (left lateral) strike-slip Altyn Tagh fault (Tapponnier et al., 2003). The East Kunlun orogenic belt merges with the Qinling orogenic belt in the east (Fig. 1). To the north occur the Qaidam and Qilian micro-continental blocks. The suture between these two blocks is marked by a 441–457 Ma eclogite belt (Song et al., 2006; Xiong et al., 2012). Farther to the north, another Paleozoic tectonic suture, the North Qilian orogenic belt, occurs between the Qilian block and the Alxa block (Fig. 2a). The U–Pb ages of zircon crystals from the eclogites in the North Qilian orogenic belt are ~464 Ma (Song

et al., 2006), similar to the ages of the eclogites in the suture between the Qilian and Qaidam blocks (Fig. 2a).

The East Kunlun orogenic belt is further divided into the South and North Kunlun orogenic belts along the central Kunlun regional fault in the literature (Fig. 2a). Abundant granitoids with zircon U–Pb ages varying from 391 Ma to 432 Ma (J. Zhang et al., 2014; Meng et al., 2013b; Wang et al., 2013) are present in the North Kunlun orogenic belt (Fig. 2b). The granitoids intruded both Precambrian metamorphic rocks and Paleozoic volcanic–sedimentary strata. Triassic sedimentary rocks are present in several small areas. Eclogites in the eastern part have a zircon U–Pb age of ~428 Ma (Meng et al., 2013a). Several ophiolite complexes with zircon U–Pb ages varying from 467 Ma to 518 Ma (Bian et al., 2004; Meng et al., 2013b; Yang et al., 1996) are present within the North Kunlun orogenic belt (Fig. 2b). The basaltic rocks of these ophiolite complexes show typical MORB geochemical signatures (Bian et al., 2004; Meng et al., 2013b; Yang et al., 1996). This, together with the presence of the 411–428 Ma eclogites in this belt (Fig. 2a, b), suggests that the North Kunlun orogenic belt was an active arc system until Late Silurian.

Several mafic–ultramafic intrusions are present in the Xiarihamu area but to date only the No. 1 intrusion has been found to contain significant Ni–Cu sulfide mineralization (Fig. 3). The mineralization potential in the other intrusions is still under investigation. The No. 1 intrusion and the associated Ni–Cu sulfide deposit are referred to as the Xiarihamu intrusion and deposit, respectively, in the literature (S.-J. Li et al., 2012).

3. Lithology and petrography

The Xiarihamu mafic–ultramafic complex consists of an older gabbroic intrusion and a younger ultramafic body intruding the gabbroic intrusion (Fig. 3). Country rocks hosting the complex are dominated by Precambrian metamorphic rocks such as biotite schist, quartz-rich schist, quartzite, marble and gneiss (S.-J. Li et al., 2012). Small diorite and granite dikes cut the Xiarihamu mafic–ultramafic intrusive rocks in places. S.-J. Li et al. (2012) and Ao et al. (2014) reported zircon U–Pb ages varying from 382 Ma to 393 Ma for the diorite dikes. No age data for the granite dikes that cut the Xiarihamu mafic–ultramafic complex are available, but a granite pluton close to the complex to the north gives a zircon U–Pb age of 391.1 ± 1.4 Ma (Wang et al., 2013).

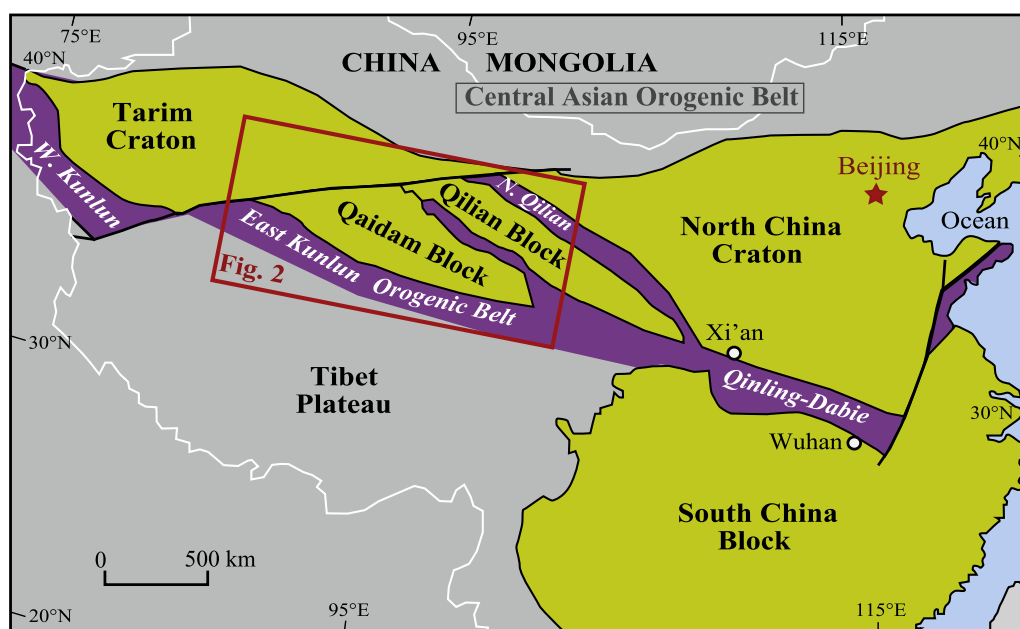


Fig. 1. Schematic map showing major continental blocks and orogenic belts in NW China.

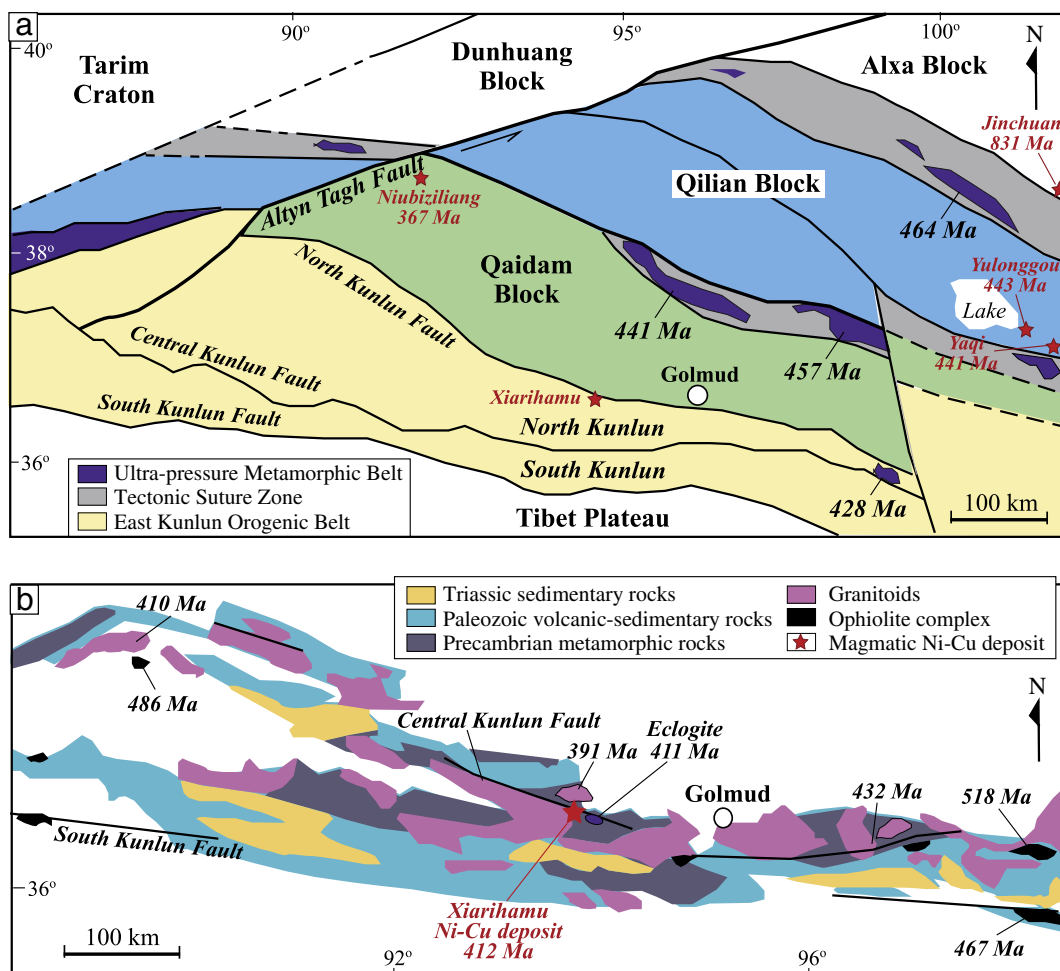


Fig. 2. Schematic maps showing the tectonic units of the northern Tibet–Qinghai plateau (a) and simplified geology of the North Kunlun arc terrane (b). The zircon U–Pb isotope age data are from Bian et al. (2004), Ling et al. (2014), Meng et al. (2013a, b), Qi et al. (2014), Song et al. (2006), Wang et al. (2013), Xiong et al. (2012), Zhang et al. (2010, 2014a, b), and this study.

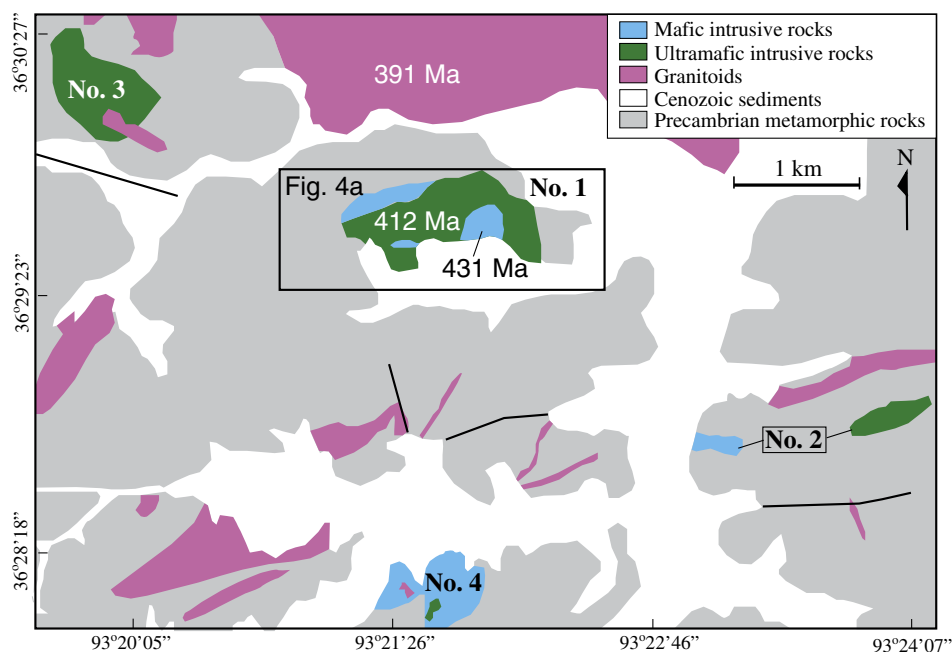


Fig. 3. Simplified geological map of the Xiarihamu region. The zircon U–Pb isotope age is from Wang et al. (2013) and this study.

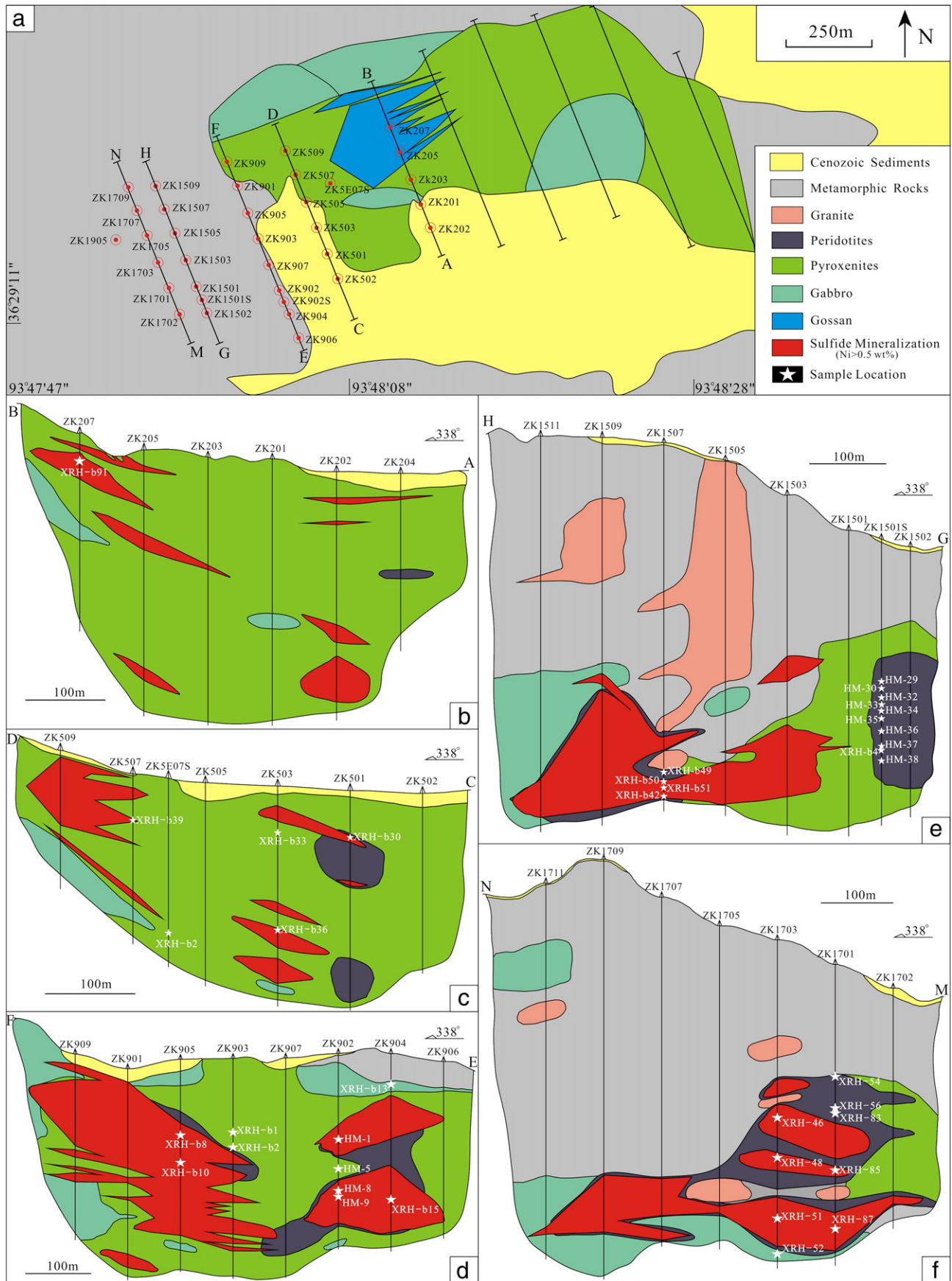


Fig. 4. Map and cross sections for the Xiarihamu magmatic sulfide deposit.

The surface-exposed area of the Xiarihamu mafic–ultramafic complex is ~0.5 km wide and ~1.6 km long (Fig. 4a). The southern part of the complex is covered by Cenozoic sediments. As shown in several selected cross sections (Fig. 4b–f), the Xiarihamu mafic–ultramafic complex gently plunges to the west. Its west end is ~500 m below Precambrian metamorphic rocks (Fig. 4f). The western part of the Xiarihamu mafic–ultramafic complex is predominantly composed of ultramafic rocks that occur continuously to form a wedge-shaped intrusive body (Fig. 4c–f). This wedge-shaped body becomes thinner downward, generally dipping to the west at 20–30°. The upper part of the ultramafic intrusion has been partially eroded by weathering (Fig. 4b, c). Its middle and lower parts are covered by older gabbroic and metamorphic rocks (Fig. 4d–f). The total thicknesses of the middle and lower parts of the ultramafic body vary from ~450 m to <150 m. Small, irregular blocks of older gabbroic rocks are preserved in places within the ultramafic intrusion (Fig. 4b–d).

Data from the on-going drilling campaign indicate that the Xiarihamu magmatic sulfide deposit contains approximately 100 million metric tons of sulfide ores with average grades of 0.8 wt.% Ni and 0.1 wt.% Cu, calculated based on an arbitrary cut-off grade of 0.6 wt.% for Ni (about three times the average Ni content of olivine in the host rocks). Massive sulfide (>80 vol.%) (Fig. 5a), heavily disseminated sulfide (>15%) (Fig. 5b, c), patchy sulfide (5–15%) (Fig. 5d, e) and weakly disseminated sulfide (<5%) (Fig. 5f) are the most important types of mineralization. The sulfide-bearing zones occur generally concordant with the lithological structure of the ultramafic body (Fig. 4b–f). The thickness and length of the heavily disseminated sulfide-mineralized zones are up to ~100 m and ~200 m, respectively. No significant sulfide mineralization was intercepted by drilling east of line #2 where the only gossan for the deposit is located (Fig. 4a).

Drilling results show that the western part of the Xiarihamu mafic–ultramafic complex is dominated by ultramafic rocks whereas the

eastern part of the complex is dominated by gabbros. The thicknesses of the western part of the complex vary between ~500 m and ~200 m, thinning toward the east. The ultramafic rocks are dominated by pyroxenites in the east and by peridotites in the west.

From drilling line #2 to #5 (Fig. 4a), the sulfide-bearing zones occur entirely within the ultramafic body, with gradational contact with the host ultramafic rocks. The sulfide-bearing zones in this area are relatively small and isolated, with gently inward dipping overall (Fig. 4b–C). Farther to the west, the sizes of the sulfide-bearing zones increase while the total number decreases (Fig. 4c–f). In the westernmost part of the ultramafic body, some of the sulfide-bearing zones become interconnected and the overall orientation of the mineralization zones changes to sub-horizontal. Some of the sulfide-bearing zones in this area are in direct contact with the country rocks such as gabbros and metamorphic rocks (Fig. 4e, f).

Based on the IUGS-recommended classification for ultramafic rocks, the most important rock types in the Xiarihamu ultramafic body are lherzolite and olivine websterite. Dunite, websterite and orthopyroxenite are of minor importance. Lherzolite and dunite belong to peridotites; orthopyroxenite, websterite and olivine websterite belong to pyroxenites. As shown in the selected cross sections (Fig. 4b–f), the proportion of peridotites generally increases toward the thinner part of the wedge-shaped Xiarihamu ultramafic body, coinciding with the direction of increasing sulfide mineralization intensity in the system.

Based on the cumulus terminology for mafic–ultramafic rocks proposed by Irvine (1982), the Xiarihamu ultramafic rocks are olivine or pyroxene cumulates. In the Xiarihamu ultramafic rocks, the ratio of orthopyroxene to clinopyroxene is about 4 to 1; plagioclase is generally rare but up to ~8 vol.% is also present in a few places (Fig. 6a). In the Xiarihamu olivine cumulates, such as dunite, lherzolite and olivine websterite, orthopyroxene occurs as large oikocrysts enclosing small olivine inclusions (Fig. 6b), minor clinopyroxene and hornblende fill in the interstitial spaces or occur as large oikocrysts enclosing relatively

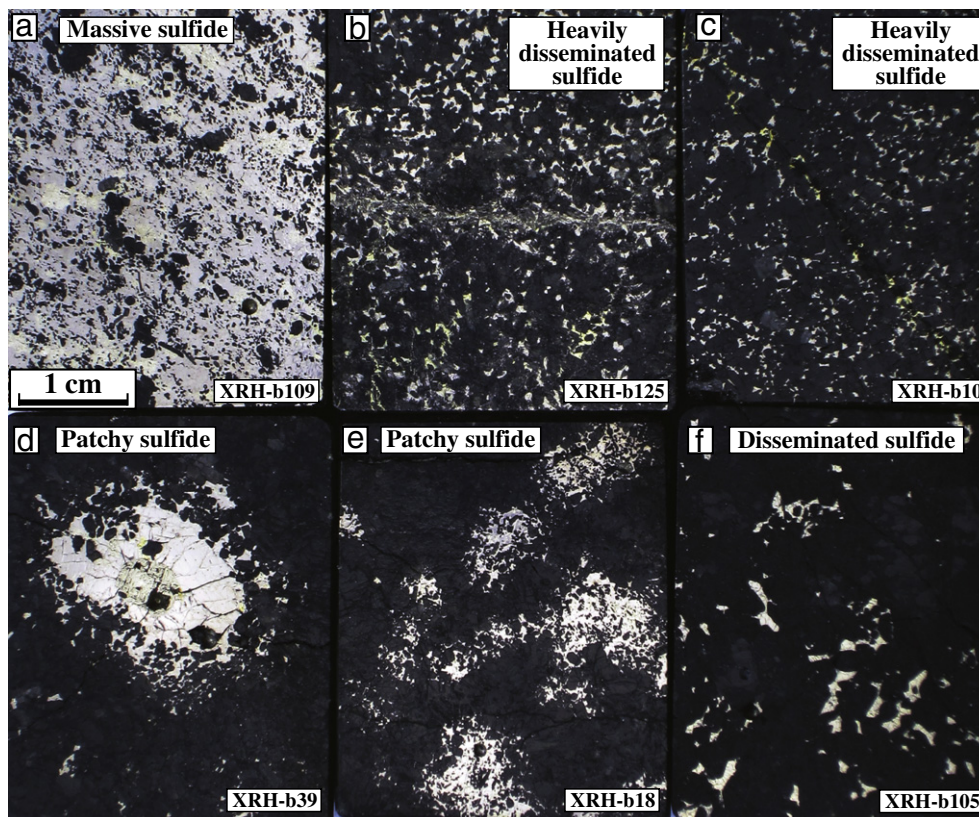


Fig. 5. Photographs of representative sulfide-mineralized samples from the Xiarihamu deposit.

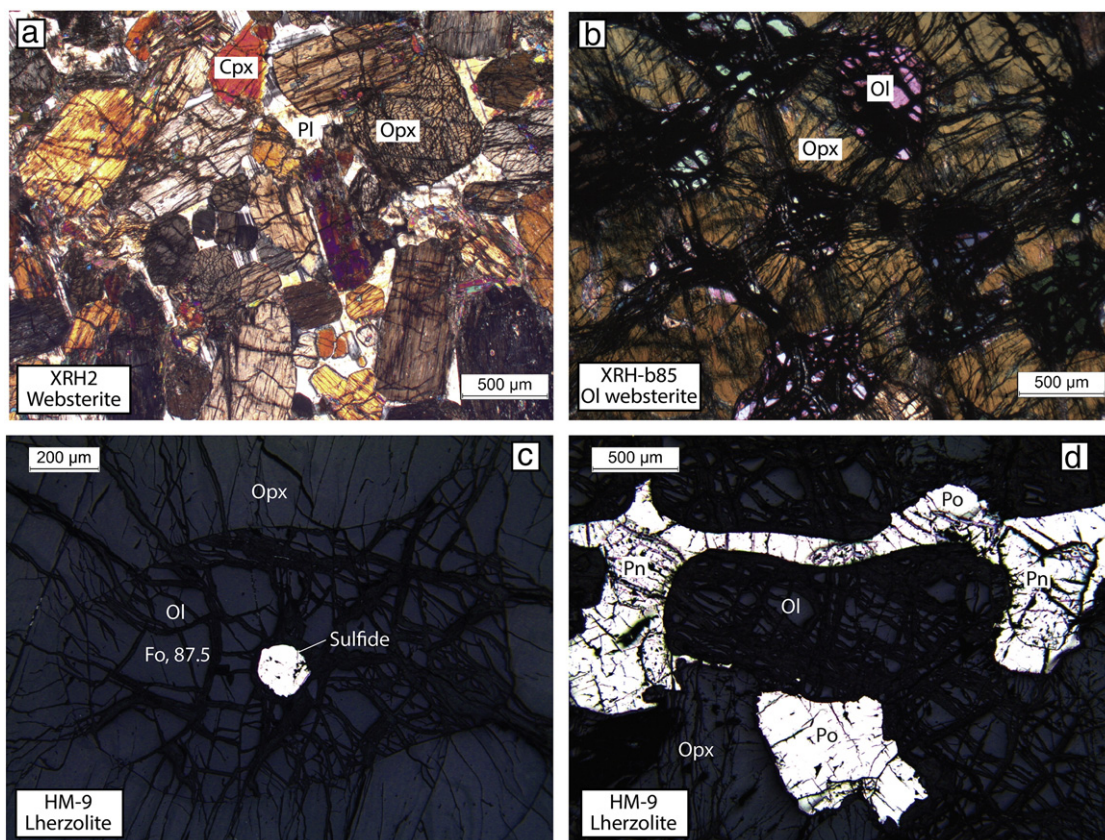


Fig. 6. Microphotographs of websterite (a), olivine websterite (b), sulfide inclusion in olivine (c), and interstitial sulfide in an olivine cumulate rock (d). Cpx = clinopyroxene, Ol = olivine, Opx = orthopyroxene, Pl = plagioclase, Pn = pentlandite, Po = pyrrhotite.

small olivine and orthopyroxene crystals. Small, rounded sulfide inclusions enclosed in olivine crystals (Fig. 6c) and in orthopyroxene crystals are common in the Xiarihamu ultramafic rocks. In disseminated sulfide samples, the sulfide assemblages (pyrrhotite + pentlandite + chalcopyrite) occur in the interstitial spaces of olivine or pyroxene cumulates (Fig. 6d).

Variable degrees of hydrothermal alteration are present in the Xiarihamu ultramafic rocks. Olivine is partially altered to serpentine; orthopyroxene is variably replaced by talc; clinopyroxene is partially altered to actinolite; plagioclase is partially replaced by clinozoisite plus albite. Chlorite and secondary magnetite occur in the micro-fractures of base-metal sulfides. In the areas not affected by faulting and deformation, alteration is weak (<30% collectively) but pervasive. In the areas affected by faulting and deformation, alteration is far more intensive, up to 100% in places.

4. Sampling and analytical methods

The samples used in this study were collected from drill cores in the western part of the Xiarihamu mafic–ultramafic complex where the best sulfide mineralization occurs (Fig. 4). Different types of sulfide mineralization and host rocks were systematically sampled. Highly altered and deformed intervals were avoided. Zircon crystals separated from a websterite sample (XRH2), a plagioclase-bearing olivine websterite sample (XRH4), and a gabbro sample (XRH6) were used for U–Pb isotope age determination. The websterite sample is from drill core ZK5E07S (N36°29'19", E92°48'07") at the depth of 347 m from the surface. The plagioclase-bearing websterite sample is from drill core ZK1501S (N36°29'13", E92°47'59") at the depth of 340 m from the surface. The gabbro sample is from drill core ZK1905 (N36°29'31", E93°36'55") at the depth of 430 m from the surface. The locations of the sampled drill cores are given in Fig. 4a. Euhedral, colorless,

transparent and un-deformed zircon crystals separated from the samples were mounted in two epoxy resin discs and then polished, cleaned and carbon coated. The cathodoluminescence and back-scattered electron images were used together to select the grains that have simple zoning patterns for dating and Hf isotope analysis.

Zircon U–Pb–Hf isotope analysis was performed using a Neptune ICP–MS equipped with a New Wave UP 213 laser-ablation sampling system in the MRL Key Laboratory of Metallogeny and Mineral Assessment, Institute of Mineral Resources, Chinese Academy of Geological Sciences, Beijing, China. The zircon GJ1 standard with the U–Th–Pb isotope ratios recommended by Jackson et al. (2004) and the Hf isotope ratio given by Morel et al. (2008) was used for calibration. The zircon Plesovice standard (337.13 ± 0.37 Ma, Sláma et al., 2008) was used as a reference. Data reduction was performed using the ICPMSDataCal program by Liu et al. (2010). Plotting and age calculations are from Isoplot/Ex 3.00 (Ludwig, 2003).

Mineral compositions were determined by wavelength dispersive X-ray analysis using a CAMECA SX50 electron microprobe at Indiana University. The analytical conditions for major and minor elements were 15 kV, 20 nA beam current, 1 μm beam size, and peak-counting time of 20 s. Nickel and Ca in olivine were analyzed using a beam current of 100 nA and a peak-counting time of 50 s. The detection limits for Ni and Ca under such conditions are 100 ppm and 50 ppm, respectively.

Whole-rock major element compositions were determined by acid digestion followed by analysis by inductively coupled plasma-optical emission spectrometry (ICP–OES) using a PerkinElmer Optima8300 machine in the National Research Center of Geoanalysis in Beijing, China. Whole-rock trace element abundances were determined by acid digestion at elevated pressure in thermally-heated steel-jacketed Teflon bombs followed by analysis by inductively coupled plasma-mass spectrometry (ICP–MS) using a Thermo Elemental X-series machine in

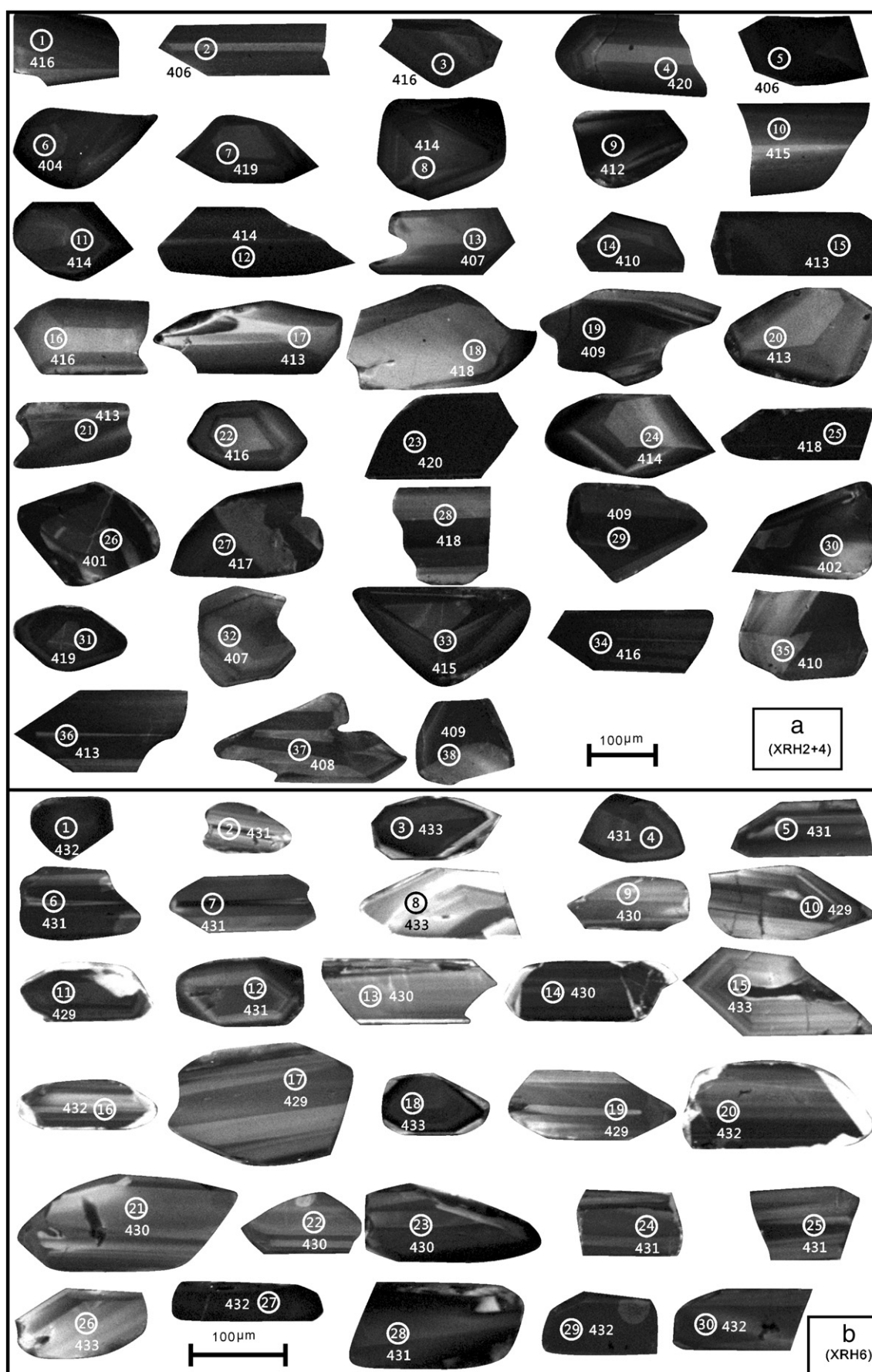


Fig. 7. Cathodoluminescence images of selected zircon crystals from Xiarihamu ultramafic rocks (a) and gabbro (b). Circles are laser beam targets. Numbers are $^{206}\text{Pb}/^{238}\text{U}$ ages (Ma).

Table 1

Concentrations of U, Th and Pb, and U–Pb isotopes of zircon from the Xiarihamu mafic–ultramafic complex.

Point number	Sample	Pb ppm	Th ppm	U ppm	Th/U	²⁰⁷ Pb/ ²⁰⁶ Pb	1σ	²⁰⁶ Pb/ ²³⁸ U	1σ	²⁰⁷ Pb/ ²³⁵ U	1σ	$t_{206/238}$ (Ma)	1σ
<i>Ultramafic rocks</i>													
1	XRH2-1	355	262	28	9.19	0.05555	0.00226	0.51008	0.02127	0.06663	0.00104	415.8	6.3
2	XRH2-2	455	334	36	9.36	0.05624	0.00267	0.49289	0.02262	0.06500	0.00177	406.0	10.7
3	XRH2-3	623	326	47	7.00	0.05501	0.00203	0.50653	0.02180	0.06661	0.00148	415.7	8.9
4	XRH2-4	256	175	20	8.81	0.05556	0.00291	0.51330	0.02816	0.06730	0.00167	419.9	10.1
5	XRH2-5	692	324	50	6.51	0.05576	0.00216	0.49681	0.01885	0.06506	0.00118	406.3	7.1
6	XRH2-6	437	244	32	7.61	0.05546	0.00238	0.49170	0.02052	0.06472	0.00137	404.2	8.3
7	XRH2-7	290	182	22	8.15	0.05573	0.00487	0.51791	0.04871	0.06717	0.00194	419.1	11.7
8	XRH2-8	227	119	16	7.26	0.05543	0.00252	0.50610	0.02293	0.06635	0.00112	414.2	6.8
9	XRH2-9	1226	1142	99	11.58	0.05639	0.00653	0.51447	0.05618	0.06599	0.00125	411.9	7.6
10	XRH2-10	200	155	15	10.08	0.05600	0.00328	0.51130	0.02872	0.06650	0.00102	415.0	6.1
11	XRH2-11	55	347	803	0.43	0.05767	0.00327	0.52998	0.03261	0.06624	0.00115	413.5	7.0
12	XRH2-12	126	1489	1686	0.88	0.05520	0.00188	0.50572	0.01676	0.06629	0.00094	413.8	5.7
13	XRH2-13	18	169	267	0.63	0.05486	0.00355	0.49638	0.03442	0.06518	0.00149	407.0	9.0
14	XRH2-14	19	171	279	0.61	0.05471	0.00448	0.49757	0.04040	0.06559	0.00162	409.5	9.8
15	XRH2-15	56	526	804	0.65	0.05753	0.00193	0.52543	0.01743	0.06620	0.00087	413.2	5.2
16	XRH2-17	10	99	148	0.67	0.05725	0.00460	0.51449	0.03777	0.06665	0.00127	416.0	7.7
17	XRH2-18	18	190	260	0.73	0.05603	0.00504	0.51336	0.04811	0.06609	0.00182	412.6	11.0
18	XRH4-1	15	102	234	0.43	0.05572	0.00336	0.51431	0.03087	0.06703	0.00188	418.2	11.4
19	XRH4-3	65	767	965	0.80	0.05831	0.00527	0.52925	0.04867	0.06550	0.00225	409.0	13.6
20	XRH4-4	16	131	253	0.52	0.05651	0.00389	0.51294	0.03612	0.06615	0.00200	412.9	12.1
21	XRH4-5	48	533	728	0.73	0.05807	0.00265	0.52924	0.02510	0.06612	0.00149	412.7	9.0
22	XRH4-7	15	118	247	0.48	0.05564	0.00854	0.50678	0.07633	0.06664	0.00285	415.9	17.2
23	XRH4-9	94	1165	1414	0.82	0.05722	0.00248	0.53039	0.02422	0.06730	0.00140	419.9	8.5
24	XRH4-10	22	249	371	0.67	0.05591	0.00271	0.50599	0.02413	0.06625	0.00126	413.5	7.6
25	XRH4-11	89	1334	1334	1.00	0.05678	0.00204	0.52307	0.01871	0.06698	0.00129	417.9	7.8
26	XRH4-12	28	306	471	0.65	0.05592	0.00267	0.48780	0.01944	0.06417	0.00091	400.9	5.5
27	XRH4-13	32	250	569	0.44	0.05612	0.00352	0.51963	0.03632	0.06682	0.00166	416.9	10.0
28	XRH4-14	38	551	619	0.89	0.05657	0.00340	0.52255	0.03081	0.06699	0.00159	418.0	9.6
29	XRH4-15	55	898	906	0.99	0.05522	0.00160	0.49963	0.01482	0.06557	0.00091	409.4	5.5
30	XRH4-16	42	424	785	0.54	0.05518	0.00277	0.48834	0.02357	0.06433	0.00111	401.9	6.7
31	XRH4-17	37	439	646	0.68	0.05760	0.00361	0.53590	0.04182	0.06713	0.00216	418.8	13.0
32	XRH4-18	17	148	331	0.45	0.05554	0.00211	0.50086	0.02009	0.06516	0.00083	406.9	5.0
33	XRH4-20	31	353	569	0.62	0.05512	0.00215	0.50467	0.01936	0.06645	0.00094	414.7	5.7
34	XRH4-21	71	1184	1231	0.96	0.05720	0.00347	0.52977	0.04327	0.06661	0.00174	415.7	10.5
35	XRH4-22	11	111	225	0.49	0.05562	0.00340	0.49963	0.03088	0.06565	0.00118	409.9	7.2
36	XRH4-23	47	788	871	0.90	0.05507	0.00182	0.50411	0.01695	0.06612	0.00092	412.7	5.6
37	XRH4-24	21	310	405	0.77	0.05433	0.00209	0.50489	0.03135	0.06537	0.00123	408.2	7.5
38	XRH4-25	32	395	655	0.60	0.05423	0.00214	0.49189	0.01811	0.06551	0.00076	409.0	4.6
<i>Gabbro</i>													
1	XRH6-1	102	439	1540	0.29	0.05917	0.00207	0.06936	0.00134	0.56778	0.02421	432.3	8.1
2	XRH6-2	26	74	408	0.18	0.05594	0.00221	0.06910	0.00145	0.52716	0.02086	430.7	8.7
3	XRH6-3	65	290	990	0.29	0.05726	0.00129	0.06946	0.00079	0.54840	0.01288	432.9	4.8
4	XRH6-4	52	197	800	0.25	0.05645	0.00156	0.06921	0.00068	0.53813	0.01449	431.4	4.1
5	XRH6-5	44	411	610	0.68	0.05909	0.00218	0.06914	0.00093	0.56311	0.02022	431.0	5.6
6	XRH6-6	54	466	765	0.61	0.05587	0.00186	0.06908	0.00084	0.53231	0.01738	430.6	5.1
7	XRH6-7	69	766	917	0.84	0.05585	0.00161	0.06920	0.00068	0.53369	0.01550	431.3	4.1
8	XRH6-8	12	80	179	0.45	0.05664	0.00332	0.06941	0.00134	0.53930	0.02990	432.6	8.1
9	XRH6-9	35	257	500	0.51	0.05951	0.00366	0.06900	0.00169	0.56318	0.03298	430.1	10.2
10	XRH6-10	49	376	697	0.54	0.05616	0.00191	0.06885	0.00098	0.53262	0.01854	429.2	5.9
11	XRH6-11	65	265	1005	0.26	0.05597	0.00141	0.06888	0.00087	0.53131	0.01418	429.4	5.3
12	XRH6-12	44	304	638	0.48	0.05716	0.00165	0.06914	0.00085	0.54245	0.01531	431.0	5.1
13	XRH6-13	23	100	337	0.30	0.05916	0.00248	0.06904	0.00085	0.56075	0.02268	430.4	5.1
14	XRH6-14	76	615	1072	0.57	0.05638	0.00136	0.06903	0.00077	0.53554	0.01261	430.3	4.7
15	XRH6-15	31	184	444	0.42	0.05825	0.00207	0.06944	0.00093	0.55525	0.01942	432.8	5.6
16	XRH6-16	45	328	634	0.52	0.05632	0.00203	0.06932	0.00093	0.53923	0.01962	432.0	5.6
17	XRH6-17	34	255	477	0.54	0.05697	0.00506	0.06875	0.00263	0.53306	0.03811	428.6	15.9
18	XRH6-18	68	297	999	0.30	0.05711	0.00169	0.06954	0.00073	0.54624	0.01536	433.4	4.4
19	XRH6-19	30	256	413	0.62	0.05463	0.00244	0.06877	0.00103	0.51940	0.02382	428.7	6.2
20	XRH6-20	33	114	497	0.23	0.06133	0.00275	0.06937	0.00090	0.58290	0.02423	432.4	5.5
21	XRH6-21	40	276	573	0.48	0.05531	0.00304	0.06898	0.00129	0.52232	0.02589	430.0	7.8
22	XRH6-22	31	247	459	0.54	0.05362	0.00211	0.06892	0.00151	0.50983	0.02162	429.6	9.1
23	XRH6-23	71	598	1013	0.59	0.05399	0.00180	0.06893	0.00145	0.51629	0.02156	429.7	8.7
24	XRH6-24	45	286	656	0.44	0.05587	0.00233	0.06906	0.00147	0.52869	0.02151	430.5	8.9
25	XRH6-25	41	354	556	0.64	0.05421	0.00241	0.06908	0.00130	0.51635	0.02433	430.6	7.8
26	XRH6-26	27	161	396	0.41	0.05578	0.00220	0.06946	0.00081	0.53355	0.02083	432.9	4.9
27	XRH6-27	152	667	2288	0.29	0.05684	0.00139	0.06932	0.00116	0.54421	0.01369	432.0	7.0
28	XRH6-28	71	423	1063	0.40	0.05431	0.00136	0.06917	0.00108	0.52026	0.01485	431.1	6.5
29	XRH6-29	113	569	1653	0.34	0.05560	0.00128	0.06930	0.00104	0.53233	0.01321	431.9	6.2
30	XRH6-30	58	318	824	0.39	0.05653	0.00175	0.06930	0.00115	0.54250	0.01873	432.0	6.9

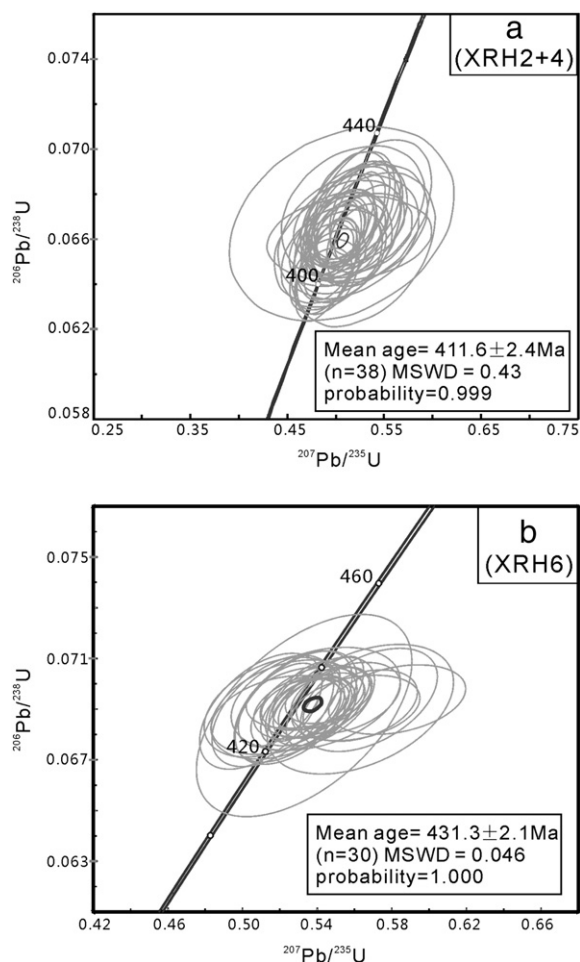


Fig. 8. Zircon U–Pb isotope Concordia plots for the Xiarihamu ultramafic rocks (a) and gabbro (b).

the National Research Center of Geoanalysis in Beijing, China. The concentrations of sulfur in whole rocks were determined by Combustion-Iodometric Titration at the Institute of Geology and Mineral Resources, Xi'an, China.

Sulfur isotope analysis was carried out at Indiana University using the continuous-flow method described in Studley et al. (2002). Coarse sulfide minerals such as pentlandite, chalcopyrite, and pyrrhotite were drilled from polished sections using a 0.75-mm carbide bit. Between 0.1 and 0.2 mg of sulfide powder was placed in tin cups with approximately 1.5 to 2 mg vanadium pentoxide (V_2O_5). Samples were prepared in an elemental analyzer by flash combustion at 1800 °C with a reactor column temperature of 1010 °C. Measurements of produced SO_2 were made using a Finnigan Delta V stable isotope ratio mass spectrometer, with results reported in per mil delta notation relative to Vienna-Canyon Diablo Troilite (VCDT). Analytical uncertainty was less than $\pm 0.05\%$, and sample reproducibility was within $\pm 0.2\%$. Sulfide standards used were IAEA-S1, IAEA-S2, IAEA-S3, with values of -0.3 , 21.7 , and -31.3% respectively, on the SO_2 scale.

5. Analytical results

5.1. Zircon U–Pb isotope age and Hf isotopes

The cathodoluminescence images of the selected zircon crystals from the Xiarihamu gabbro and ultramafic rock samples are shown in Fig. 7. The analytical results of the selected zircon crystals are listed in Table 1. The Concordia plots of the analyses are shown in Fig. 8a and b, respectively. The weighted mean zircon U–Pb isotope age for the

Xiarihamu ultramafic rocks is 411 ± 2.4 Ma (Fig. 7a). The weighted mean zircon U–Pb isotope age for the associated gabbro is 431.3 ± 2.1 Ma (Fig. 7b). The results show that the Xiarihamu ultramafic body is ~ 20 Ma younger than the associated gabbro. To our knowledge such a large age gap between mafic and ultramafic rocks has never been reported for layered intrusions formed in intraplate settings but is very common for the Alaskan-type mafic–ultramafic complexes such as the Duke Island complex in southeastern Alaska in the United States (Gehrels et al., 1987; Saleeby, 1992).

The Lu–Hf isotopes of the selected zircon crystals from the Xiarihamu ultramafic rocks are listed in Table 2. The calculated $\epsilon_{Hf}(t)$ values of these zircon crystals are all positive, varying from 1 to 5. A comparison of these values with those for a depleted mantle and a potential contaminant, the Precambrian gneiss which is exposed in the eastern part of the Qaidam orogenic belt at Dulan (data from Xiong et al., 2012), reveals that up to 15% contamination is permitted for the parental magma of the Xiarihamu ultramafic rocks (Fig. 9). Precambrian gneissic rocks have been intercepted by drilling at Xiarihamu but their Hf isotope compositions are not available currently.

5.2. Mineral compositions

The compositions of important rock-forming minerals such as olivine, pyroxenes, plagioclase, hornblende and Cr-spinel are given as supplementary data (Table S1). Cr-spinel is rare in the Xiarihamu ultramafic intrusion. Small Cr-spinel inclusions enclosed in olivine in two lherzolite samples contain 34–44 wt.% Al_2O_3 , 20–31 wt.% Cr_2O_3 and 11–13 wt.% MgO, with $Cr^\#$ [$Cr/(Cr + Al)$, molar] varying between 0.24 and 0.37. The $Cr^\#$ of Cr-spinel in the Xiarihamu ultramafic intrusion are significantly lower than those of Cr-spinel in the ultramafic rocks of the Duke Island complex in southern Alaska, a typical Alaskan-type mafic–ultramafic complex in an arc setting (Thakurta et al., 2008).

The An contents of plagioclase in an olivine websterite sample from the Xiarihamu ultramafic intrusion are between 65 and 69 mol%. The En contents of clinopyroxene in two lherzolite samples are 51–56 mol%. The En contents of clinopyroxene in two olivine websterite samples are slightly lower, varying from 47 to 52 mol%. The $Mg^\#$ [$100 Mg/(Mg + Fe)$, molar] of orthopyroxene in lherzolite, olivine websterite and websterite are similar, varying from 85 to 90. These values are similar to those of orthopyroxene in ultramafic rocks formed from subduction-related boninitic basalt such as the Erbutu sulfide-bearing ultramafic intrusion in the Inner Mongolia, China (Peng et al., 2013).

5.3. Olivine systematics

Olivine in the Xiarihamu ultramafic intrusion contains 84–90 mol% Fo, 500–2000 ppm Mn and <700 ppm Ca (Fig. 10). High-Mg olivine (i.e., Fo > 89 mol%) with Ca-depletion (i.e., Ca < 1000 ppm) is most common in mantle peridotites (Simkin and Smith, 1970), arc basalts (Kamenetsky et al., 2006) and Alaskan-type mafic–ultramafic complex such as the Duke Island complex in southern Alaska (C. Li et al., 2012), but extremely rare in layered intrusions in intraplate settings (e.g., Emeishan large igneous province, Wang et al., 2014).

The stratigraphic variations of olivine Fo and Ni contents across the Xiarihamu magmatic sulfide deposit from east to west are illustrated in Figs. 11–15. The locations of the selected drill cores are shown in Fig. 4a. Fig. 11a is the profile for drill core ZK902S which is located on the drilling line #9 (Fig. 4a). The analyzed samples from this borehole are all sulfide-mineralized lherzolite, containing trace sulfide (<0.5 wt.% S), disseminated sulfide (0.5–2.5 wt.% S) or heavily disseminated sulfide (2.5–5 wt.% S). In this drill core the mineralized interval can be further divided into two zones with similar Fo contents but significantly different Ni contents in olivine. In each zone olivine shows a negative Ni–Fo correlation (Fig. 11b). The olivine Ni contents in the upper zone are about double the values in the lower zone (Fig. 11b). These two zones also show different stratigraphic variations in olivine Fo–Ni

Table 2

Hf isotopes of zircon crystals from the Xiarihamu ultramafic body (sample XRH2, XRH4).

No	$^{176}\text{Yb}/^{177}\text{Hf}$	2σ	$^{176}\text{Lu}/^{177}\text{Hf}$	2σ	$^{176}\text{Hf}/^{177}\text{Hf}$	2σ	$(^{176}\text{Hf}/^{177}\text{Hf})_i$	ϵ_{Hf}
XRH-1	0.014866	0.000111	0.000482	0.000002	0.282608	0.000016	0.282604	3.1
XRH-2	0.019239	0.000189	0.000620	0.000004	0.282612	0.000016	0.282607	3.2
XRH-3	0.011055	0.000207	0.000390	0.000006	0.282608	0.000016	0.282605	3.2
XRH-4	0.010289	0.000205	0.000359	0.000006	0.282615	0.000016	0.282612	3.4
XRH-5	0.016899	0.000072	0.000633	0.000002	0.282617	0.000017	0.282612	3.4
XRH-6	0.018579	0.000031	0.000656	0.000000	0.282581	0.000018	0.282576	2.1
XRH-7	0.015303	0.000031	0.000520	0.000001	0.282554	0.000017	0.282550	1.2
XRH-8	0.016453	0.000030	0.000584	0.000003	0.282554	0.000018	0.282550	1.2
XRH-9	0.029751	0.000158	0.001113	0.000004	0.282556	0.000022	0.282548	1.1
XRH-10	0.010714	0.000111	0.000356	0.000003	0.282580	0.000019	0.282578	2.2
XRH-11	0.016737	0.000117	0.000622	0.000005	0.282591	0.000016	0.282586	2.5
XRH-12	0.035882	0.000064	0.001231	0.000004	0.282652	0.000019	0.282643	4.5
XRH-13	0.011649	0.000211	0.000402	0.000007	0.282594	0.000016	0.282591	2.6
XRH-14	0.013310	0.000094	0.000460	0.000002	0.282605	0.000017	0.282601	3.0
XRH-15	0.030400	0.000300	0.001045	0.000007	0.282624	0.000017	0.282616	3.5
XRH-16	0.010983	0.000204	0.000374	0.000007	0.282592	0.000017	0.282589	2.6
XRH-17	0.013671	0.000303	0.000445	0.000009	0.282583	0.000016	0.282580	2.3
XRH-18	0.015166	0.000256	0.000595	0.000010	0.282600	0.000022	0.282595	2.8
XRH-19	0.022790	0.000310	0.000815	0.000010	0.282609	0.000020	0.282603	3.1
XRH-20	0.016470	0.000064	0.000591	0.000002	0.282555	0.000021	0.282551	1.2
XRH-21	0.041852	0.000274	0.001458	0.000009	0.282573	0.000022	0.282562	1.6
XRH-22	0.012499	0.000050	0.000487	0.000002	0.282600	0.000020	0.282596	2.8
XRH-23	0.044710	0.000717	0.001596	0.000024	0.282579	0.000023	0.282567	1.8
XRH-24	0.015602	0.000051	0.000589	0.000003	0.282545	0.000020	0.282541	0.9
XRH-25	0.040917	0.000359	0.001439	0.000011	0.282623	0.000019	0.282612	3.4
XRH-26	0.028247	0.000127	0.001048	0.000003	0.282612	0.000022	0.282604	3.1
XRH-27	0.026260	0.000317	0.001038	0.000012	0.282591	0.000020	0.282583	2.4
XRH-28	0.022635	0.000362	0.000830	0.000010	0.282602	0.000023	0.282596	2.8
XRH-29	0.038633	0.000414	0.001365	0.000012	0.282607	0.000027	0.282596	2.8
XRH-30	0.030934	0.000065	0.001139	0.000002	0.282570	0.000023	0.282562	1.6
XRH-31	0.027831	0.000171	0.001083	0.000009	0.282591	0.000027	0.282582	2.3
XRH-32	0.019528	0.000069	0.000758	0.000001	0.282621	0.000019	0.282615	3.5
XRH-33	0.024999	0.000191	0.000952	0.000005	0.282580	0.000023	0.282573	2.0
XRH-34	0.041281	0.000288	0.001465	0.000013	0.282590	0.000021	0.282579	2.2
XRH-35	0.015023	0.000092	0.000578	0.000005	0.282559	0.000016	0.282554	1.3
XRH-36	0.036316	0.000328	0.001296	0.000010	0.282562	0.000023	0.282552	1.3
XRH-37	0.019456	0.000116	0.000734	0.000003	0.282654	0.000019	0.282648	4.7
XRH-38	0.030595	0.000207	0.001160	0.000007	0.282650	0.000022	0.282641	4.4

ϵ_{Hf} calculated using the method of Blichert and Albarede (1997), ^{176}Lu decay constant $\lambda = 1.865 \times 10^{-11} \text{y}^{-1}$ (Soderlund et al., 2004), $t = 411.6 \text{ Ma}$.

contents (Fig. 11a). In the lower zone, olivine Fo contents decrease upward. The sense of variation is opposite in the upper zone.

Fig. 12a illustrates the stratigraphic variations of olivine Fo and Ni contents in drill core ZK1501S which is located on the drilling line #15 (Fig. 4a). The analyzed samples from this borehole are all lherzolite with minor sulfide mineralization (0.3–1 wt.% S). Based on different

olivine Fo–Ni contents, this drill core can be further divided into 4 zones (Fig. 12a). Olivine crystals from zone 2 and zone 4 collectively show a clear negative Ni–Fo correlation (Fig. 12b). The Fo contents of olivine crystals from zone 1 and zone 3 do not change much but their Ni contents vary significantly between 1200 ppm and 2500 ppm. Olivine crystals from zone 1 and zone 3 are much poorer in Ni content than those from zone 2 and zone 4.

The stratigraphic variations of olivine Fo–Ni contents in another drill core (ZK1507) on the drilling line #15 are shown in Fig. 13a. The analyzed samples from this drill core are all lherzolite. The interval we have sampled can be further divided into 3 zones based on different

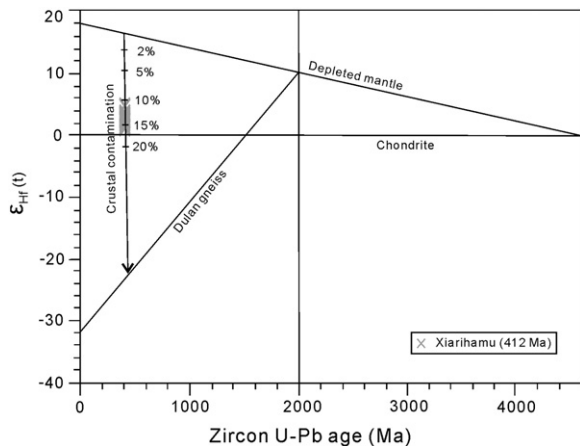


Fig. 9. Zircon ϵ_{Hf} (t) value versus zircon U–Pb isotope age for the Xiarihamu ultramafic rocks. The age of the Dulan gneiss is from Xiong et al. (2013). The Hf isotope evolution of the Dulan gneiss is calculated using the parameters for the upper crust given by Amelin et al. (1999). The average concentration of Hf (1.95 ppm) in global island arc subalkaline basalts from a public online database (<http://www.petdb.org>) is used to represent the composition of the mantle-derived magma.

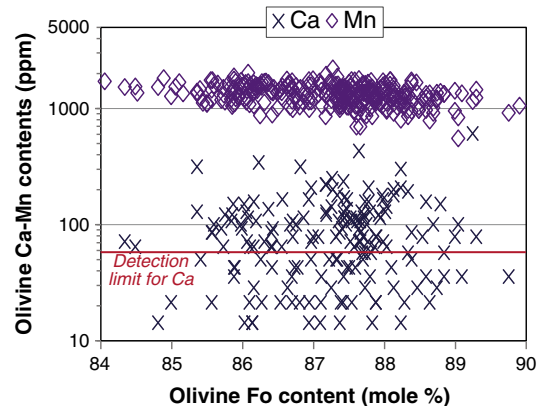


Fig. 10. The Ca, Mn and Fo contents of olivine in the Xiarihamu ultramafic rocks.

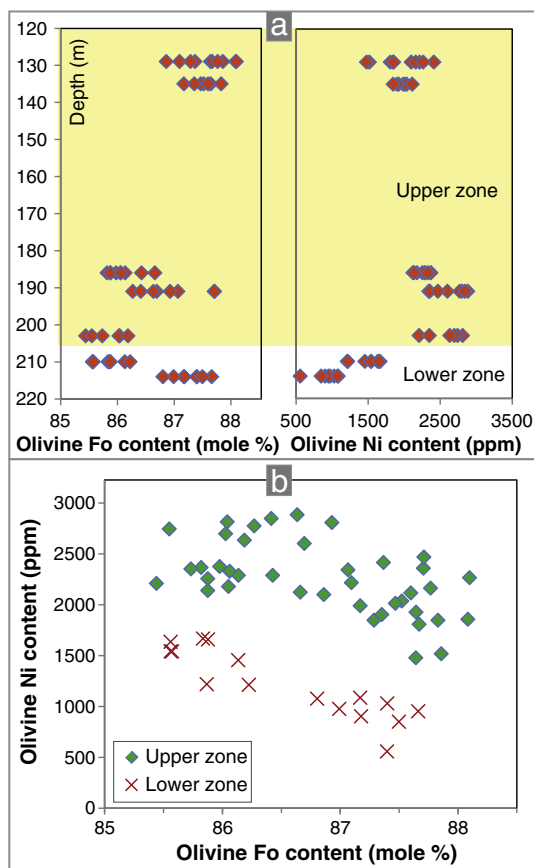


Fig. 11. Olivine stratigraphic compositional variations (a) and Ni–Fo correlations (b) for drill core ZK902S.

Fo–Ni contents in olivine. Zone 1 and zone 3 contain trace amounts of sulfide (<0.5 wt.% S); zone 2 contain weakly disseminated sulfide (0.6–2.2 wt.% S). Olivine crystals from the middle zone (i.e., zone 2) have higher Fo and lower Ni contents than those from the other two zones. Olivine inclusions enclosed in large orthopyroxene crystals in zone 1 and zone 3 collectively show a positive Ni–Fo correlation.

Fig. 14a illustrates the stratigraphic variations of olivine Fo–Ni contents in drill core ZK1701 which is located on the drilling line #17 (see Fig. 4a). Samples from this drill core are either lherzolite or olivine websterite. Zones containing trace sulfide and zones containing disseminated sulfide alternate in this drill core. Olivine inclusions enclosed in large orthopyroxene crystals in the uppermost trace sulfide zone show rapid decrease in both Fo and Ni contents upward (Fig. 14a). The same type of olivine crystals in the underlying trace sulfide zones collectively show gradual decrease in both Fo and Ni downward. In contrast, the disseminated sulfide zones collectively show upward decreasing Fo contents coupled by increasing Ni contents. As shown in Fig. 14b, olivine crystals from all disseminated sulfide zones in this drill core show a weak negative Ni–Fo correlation. Olivine inclusions enclosed in large orthopyroxene crystals from the trace sulfide zones in this drill core collectively show a clear positive correlation. Such correlation is also present in a single sample between crystals (the sample at 380 m, Fig. 14b). In this sample, Ni content in olivine decreases from ~2700 ppm down to ~2000 ppm as Fo content decreases from ~87.8 mol% to ~86.4 mol%.

5.4. Whole-rock major and trace elements

The concentrations of major and trace elements in the samples from the Xiarihamu ultramafic intrusion used in this study are given as supplementary data (Table S2). A comparison of whole-rock compositions with the compositions of major rock-forming minerals is illustrated in

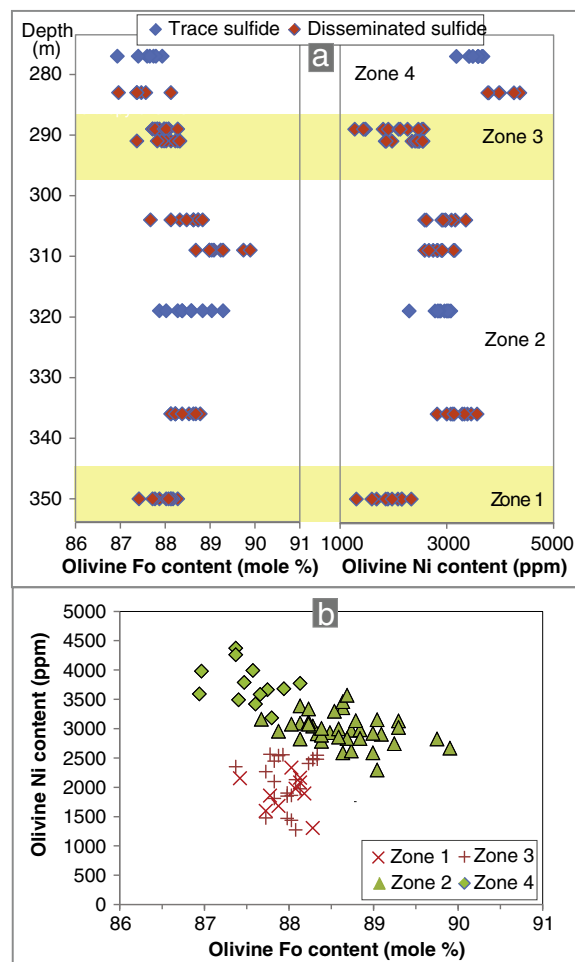


Fig. 12. Olivine stratigraphic compositional variations (a) and Ni–Fo correlations (b) for drill core ZK1501S.

Fig. 15a–c. In the comparison the whole-rock compositions were normalized to 100% on a LOI- and sulfide-free basis. The total amount of Fe contributed from the base-metal sulfides (pyrrhotite, pentlandite and chalcopyrite) in a sample was estimated from S, Ni (corrected for contribution from olivine) and Cu contents in the sample using the equation of Barnes and Lightfoot (2005). The corrected Ni values are whole-rock concentrations minus 1300 ppm. The value used for correction (i.e., 1300 ppm Ni) is close to the value for the most sulfide-poor lherzolite sample (1600 ppm S) in our collection (see Table S2). The comparison reveals that the major element compositions of dunite–lherzolite and websterite–orthopyroxenite samples from the Xiarihamu ultramafic intrusion are mainly controlled by the abundances of olivine and pyroxenes, respectively, confirming that the Xiarihamu ultramafic rocks are olivine or pyroxene cumulates. All the samples including the sulfide-poor samples have Cu/Zr ratios >1, indicating the presence of cumulus sulfide in the rocks (Li and Naldrett, 1999). Due to the presence of variable amounts of cumulus sulfide, the Cu/Zr ratios are positively correlated with S contents in the samples.

The primitive mantle-normalized immobile trace element patterns for relatively sulfide-poor samples (<1 wt.% S) from the Xiarihamu ultramafic intrusion are illustrated in Fig. 16. These samples are characterized by light REE enrichments relative to heavy REE (Fig. 9a, c) and a pronounced negative Nb anomaly relative to Th and La. Tantalum is also depleted in the samples but not shown in this diagram because its concentrations in some of the samples are below the detection limit (see Table S2). The trace element patterns of the Xiarihamu ultramafic rocks are typical for sulfide-bearing arc ultramafic cumulates such as

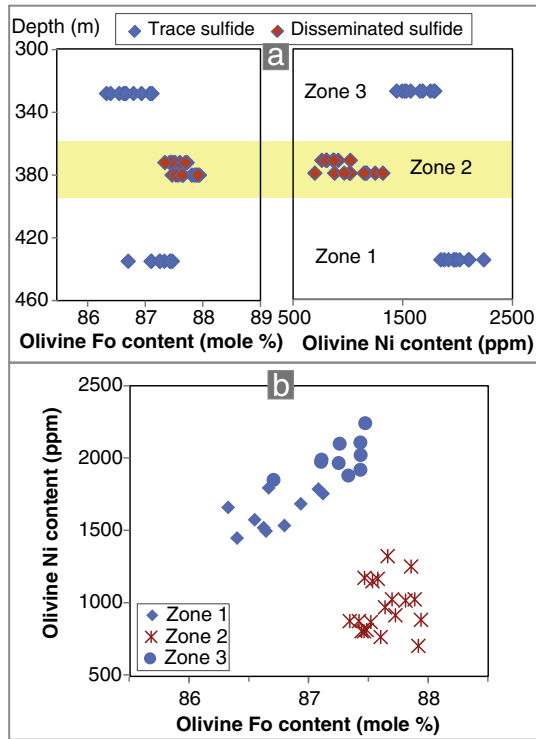


Fig. 13. Olivine stratigraphic compositional variations (a) and Ni–Fo correlations (b) for drill core ZK1507.

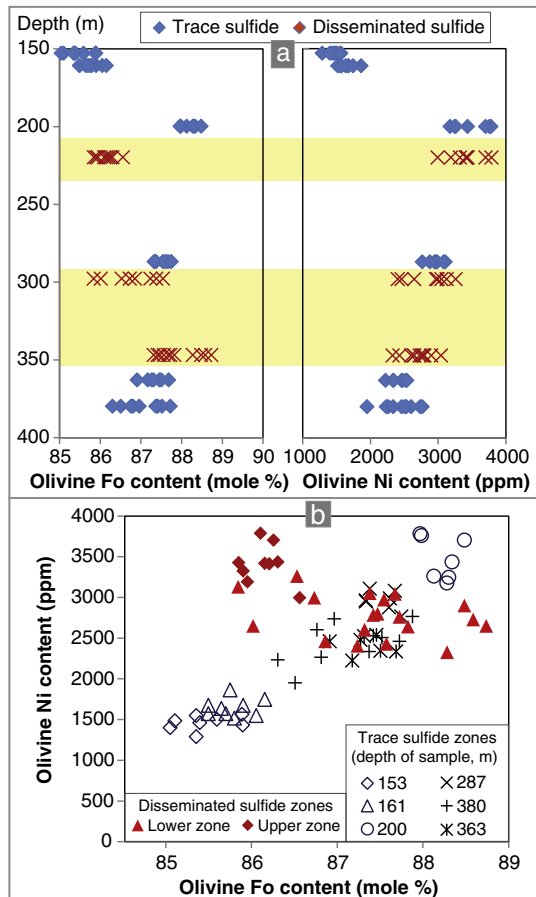


Fig. 14. Olivine stratigraphic compositional variations (a) and Ni–Fo correlations (b) for drill core ZK1701.

the Permian Erbutu sulfide-bearing ultramafic intrusion in an arc setting in Inner Mongolia, China (Peng et al., 2013).

5.5. Sulfur isotopes

Sulfur isotope compositions of sulfide separates from the Xiarihamu Ni–Cu sulfide deposit are listed in Table 3. The $\delta^{34}\text{S}$ value of pyrrhotite separates from lherzolite-hosted disseminated sulfides are from 3.5‰ to 5.7‰. The disseminated sulfides hosted in olivine websterite have $\delta^{34}\text{S}$ values varying from 3.8‰ to 4.6‰. Massive and net-textured sulfides associated with olivine websterite have similar $\delta^{34}\text{S}$ values close to 6.3‰. There are no systematic variations of $\delta^{34}\text{S}$ values among different types of sulfide mineralization associated with websterite such as massive, disseminated and patchy sulfides, and sulfide blebs. The $\delta^{34}\text{S}$ values of the various types of sulfide ores associated with websterite are all elevated, varying from 5.2 to 6.8‰.

6. Modeling and discussion

6.1. Oxidation state

The oxidation state of a sulfide-bearing magmatic system may be estimated from Ni concentration in sulfide liquid and the exchange coefficient for Fe and Ni between olivine and sulfide liquid, $K_D [(X_{\text{NiS}}/X_{\text{FeS}})_{\text{sulfide liquid}}/(X_{\text{NiS}}/X_{\text{FeS}})_{\text{olivine}}]$, molar (Brenan, 2003). Recently, Barnes et al. (2013) used available experimental dataset to calibrate this relationship. We have used the equation of Barnes et al. (2013) to estimate the oxidation state for the Xiarihamu magmatic sulfide deposit based on two sulfide-bearing lherzolite samples containing 2.6 wt.% S (sample HM-9) and 3.3 wt.% S (XRH-b48). Prior to such calculation, the contributions of Ni from olivine in the samples were corrected using the average Ni contents in olivine together with abundances of olivine in the samples; Ni tenors (concentrations in recalculated 100% sulfide) in the samples were estimated using the equation of Barnes and Lightfoot (2005). As shown in Fig. 17, the estimated oxidation state for the Xiarihamu deposit is close to QFM + 1 (i.e., f_{O_2} one log unit higher than the value for the quartz–fayalite–magnetite buffer).

It is interesting that the estimated oxidation state of the Xiarihamu deposit is more oxidized than MORB but within the range of arc basalts based on data given in Kelley and Cottrell (2012). This may be used to support the interpretation that the oxidation state of a magmatic sulfide deposit mainly reflects the source mantle conditions. However, such interpretation has limitation because crustal contamination may also change the oxidation state of magma. For examples, the more reduced Voisey's Bay deposit in Labrador, Canada was thought to be due to assimilation of graphite-rich country rocks by magma (Ripley et al., 2002) whereas the relatively more oxidized Jinchuan deposit was believed to be due to magma interaction with marble country rocks by some researchers (Lehmann et al., 2007).

6.2. Parental magma

The magma composition of a rock formed by solidification of an olivine crystal mush can be estimated from olivine and whole-rock compositions based on the exchange coefficient for Fe and Mg between olivine and magma, $K_D [(FeO/MgO)_{\text{olivine}}/(FeO/MgO)_{\text{magma}}]$ and mass balance (Li and Ripley, 2011). As shown in Figs. 6 and 16, the textures and major oxide compositions of the Xiarihamu peridotites are consistent with the products of olivine crystal mushes at magmatic temperatures. Thus, the method of Li and Ripley can be applied to estimate the liquid compositions of the Xiarihamu peridotites. Two sulfide-poor (0.32–0.36 wt.% S) lherzolite samples (HM-36, XRH-b83) containing olivine with Fo contents varying from 88 to 89.3 mol% are most suitable for this exercise. The compositions of olivine in these samples are the most primitive among the lherzolite samples containing <1 wt.% S. More primitive olivine with Fo contents as high as 89.9 mol% is

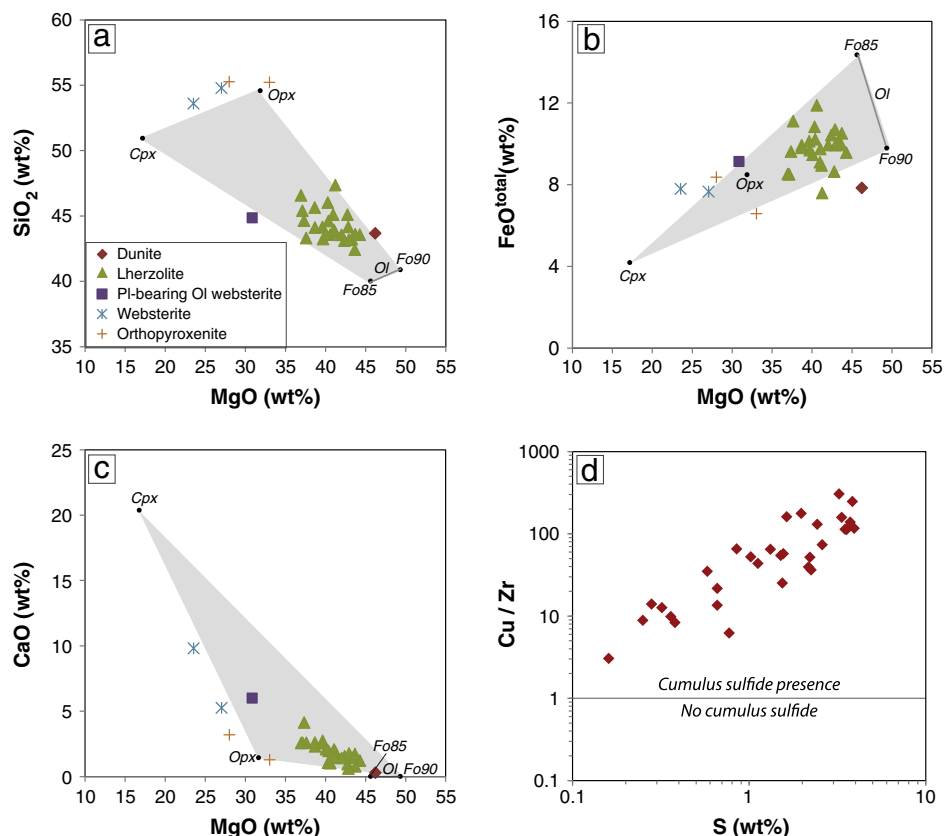


Fig. 15. Variations of whole rock compositions in the Xiarihamu ultramafic intrusion. The upper limit of Cu/Zr ratios for mafic–ultramafic rocks containing no cumulus sulfide is from Li and Naldrett (1999). Mineral abbreviations are the same as in Fig. 6.

present in several samples with S contents > 1 wt.%. However, these samples are not as useful as the sulfide-poor samples because the errors associated with correction for Fe in base-metal sulfides are larger for the samples with higher sulfide contents than the sulfide-poor samples. In our calculations whole-rock compositions were corrected for the contributions of Ni from olivine in the samples using the average Ni contents in olivine together with the abundances of olivine in the samples. This was followed by correction for Fe in the base-metal sulfides (pyrrhotite, pentlandite and chalcopyrite). The amounts of base-metal sulfides in the selected samples were estimated following the

procedures given in Li et al. (2001). The corrected whole-rock compositions were then normalized to 100% free of LOI and sulfide. The average values of the normalized compositions of the selected samples, together with the ideal composition of pure Mg–Fe olivine with Fo = 89.9 mol% and an olivine–liquid Fe–Mg exchange coefficient of 0.3 which is most suitable for basaltic magma (Roeder and Emslie, 1970) were used to determine the liquid composition. In the calculations, the ratio of ferric and ferrous Fe in whole rock was repeatedly adjusted until this ratio in the liquid is consistent with the oxidation state of QFM + 1. The liquid temperature and corresponding $\text{Fe}_2\text{O}_3/\text{FeO}$ ratio in the liquid were calculated using the MELTS program of Ghiorsso and Sack (1995). Total pressure was assumed to be 1 kb, which corresponds to ~3 km at depth. The results are given in Table 4. The estimated MgO, FeO and SiO_2 contents in the liquid are 9.79, 6.54 and 52.42 wt.%, respectively. Since the composition of the most primitive olivine in the intrusion was used in the calculations, the estimated liquid can be regarded as the best approximation for the most primitive parental magma for the intrusion. Interestingly, the composition of the estimated parental magma for the Xiarihamu ultramafic intrusion is within the range of arc boninites based on a public online database (<http://www.petdb.org>).

6.3. Timing and cause of sulfide saturation

We have evaluated the timing of sulfide saturation in the Xiarihamu magma by comparing the observed positive Fo–Ni relationship of olivine inclusions enclosed in large orthopyroxene crystals which prevented Fe–Ni exchange reaction between olivine and sulfide liquid by physical isolation (e.g., Fig. 6c) in sulfide-poor lherzolite samples containing < 0.5 wt.% S with the result for simulated olivine fractional crystallization from the estimated parental magma for the Xiarihamu lherzolites. At 1 kb total pressure and the oxidation state of QFM + 1,

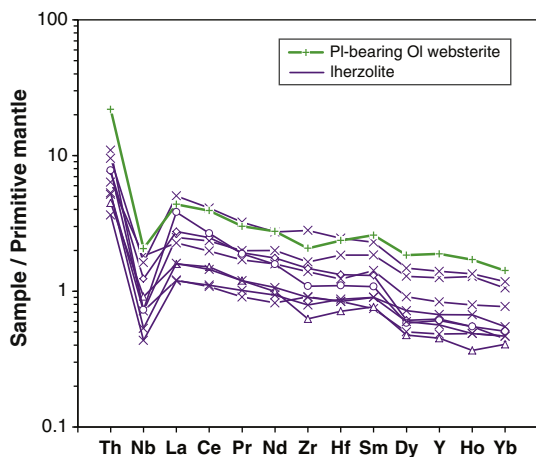


Fig. 16. Primitive mantle-normalized immobile incompatible trace element patterns for the Xiarihamu ultramafic rocks. The primitive mantle values are from Sun and McDonough (1989).

Table 3
S isotopes of the Xiarihamu magmatic Ni–Cu ore deposit.

Sample	Drill core	Depth (m)	Host rock	Sulfide texture	Mineral	$\delta^{34}\text{S}$, ‰
XRH-b11	ZK905	273	Websterite	Patchy sulfide	Pentlandite	5.62
XRH-b11	ZK905	273	Websterite	Patchy sulfide	Pyrrhotite	5.66
XRH-b97	ZK202	235	Websterite	Patchy sulfide	Pyrrhotite	6.47
XRH-b10	ZK905	152	Websterite	Disseminated sulfide	Pyrrhotite	5.17
XRH-b18	ZK909	175	Websterite	Heavily disseminated sulfide	Pyrrhotite	6.30
XRH-b30	ZK501	71	Websterite	Heavily disseminated sulfide	Pyrrhotite	6.80
XRH-b30	ZK501	71	Websterite	Heavily disseminated sulfide	Pyrrhotite	6.80
XRH-b33	ZK503	68	Websterite	Heavily disseminated sulfide	Chalcopyrote	6.41
XRH-b33	ZK503	68	Websterite	Heavily disseminated sulfide	Pyrrhotite	6.41
XRH-b39	ZK507	65	Websterite	Sulfide blebs	Pentlandite	6.06
XRH-b39	ZK507	65	Websterite	Sulfide blebs	Pyrrhotite	6.31
XRH-b39	ZK507	65	Websterite	Sulfide blebs	Pyrrhotite	5.99
XRH-b4	ZK903	214	Websterite	Massive sulfide	Pentlandite	5.75
XRH-b4	ZK903	214	Websterite	Massive sulfide	Pyrrhotite	5.98
XRH-b36	ZK503	173	Websterite	Massive sulfide	Pyrrhotite	5.92
XRH-b58	ZK1107	135	Olivine websterite	Heavily disseminated sulfide	Chalcopyrote	4.56
XRH-b58	ZK1107	135	Olivine websterite	Heavily disseminated sulfide	Pentlandite	3.99
XRH-b58	ZK1107	135	Olivine websterite	Heavily disseminated sulfide	Pyrrhotite	4.01
XRH-b125	ZK1101	100	Olivine websterite	Disseminated sulfide	Pyrrhotite	3.76
XRH-b92	ZK207	57	Olivine websterite	Net-textured sulfide	Pyrrhotite	6.40
XRH-b109	ZK011	131	Olivine websterite	Massive sulfide	Pyrrhotite	6.27
XRH-b109	ZK011	131	Olivine websterite	Massive sulfide	Pyrrhotite	6.26
XRH-b105	ZK1309	246	Lherzolite	Disseminated sulfide	Pyrrhotite	3.49

the result of simulation using the MELTS program of [Ghiorsso and Sack \(1995\)](#) shows that olivine containing ~88 mol% Fo crystallizes alone from the parental magma at 1270 °C. The Fo content of olivine decreases to ~85 mol% at 1228 °C. On further cooling the magma crystallizes orthopyroxene instead of olivine. The simulated result of fractional crystallization is remarkably similar to the observed crystallization sequence as revealed by textures (e.g., [Fig. 6b, c](#)). The total amount of olivine produced from the magma by fractional crystallization is ~4 wt.%. The average partition coefficient for Ni between olivine and magma during fractional crystallization, calculated using the equation of [Li and Ripley \(2010\)](#), is 10. Using these parameters, we have determined how Ni content in olivine varies with Fo content if olivine crystallizes alone from the magma. Our modeling results indicate that fractional crystallization of olivine alone cannot explain the observed olivine Fo–Ni relationship in the samples ([Fig. 18](#)). Concurrent sulfide segregation from the magma during olivine crystallization is required. Using a partition coefficient of 500 for Ni between sulfide liquid and magma, which is

within the range of experimental results for basaltic systems (see summary in [Naldrett, 2011](#)), our calculation shows that a mass olivine/sulfide ratio of ~40 can well explain the observed Fo–Ni relationship for the samples ([Fig. 18](#)). Our modeling results show that the Xiarihamu magma was saturated with immiscible sulfide liquid when primitive olivine containing Fo > 88 mol% was crystallizing from the magma. This is consistent with the presence of primary sulfide inclusions enclosed in olivine crystals which in turn are enclosed by poikilitic orthopyroxene crystals (e.g., [Fig. 6c](#)).

Crustal contamination and fractional crystallization are widely regarded as the most common processes involved in inducing sulfide saturation in basaltic magma (e.g., [Ripley and Li, 2013](#)). As shown above, the parental magma of the Xiarihamu magmatic sulfide deposit is rather primitive, capable of crystallizing olivine with Fo content as high as 89.9 mol%, which is close to the composition of mantle olivine. This means that the parental magma of the Xiarihamu ultramafic rocks experienced a very small degree of olivine fractional crystallization (<3 wt.%). It is highly unlikely that sulfide saturation in the Xiarihamu magma was triggered by such a small degree of olivine fractional crystallization. Addition of crustal sulfur to the magma appears to be a more viable explanation. This is consistent with higher $\delta^{34}\text{S}$ values (3.5 to 6.8‰) of the sulfide ores than the typical mantle value (0 ± 2 ‰), although the sulfur contents and isotopes in the actual contaminants are yet to be investigated.

6.4. Multiple sulfide-bearing magma pulses

Fe–Ni exchange reaction between olivine crystals and a sulfide liquid will produce a negative Fo–Ni correlation for the olivine crystals (see summary in [Li et al., 2007](#)). The sub-parallel arrays of negative Fo–Ni correlation for olivine crystals from different sulfide-mineralized zones in the Xiarihamu deposit ([Figs. 11, 12](#)) indicate different Ni contents in the sulfide liquids in the different zones. Such variation cannot be explained by in situ fractional crystallization of olivine from a single magma because the ranges of olivine Fo contents in the different zones are similar. A more plausible explanation is that these separate sulfide-bearing zones in the deposit formed from multiple pluses of magma containing immiscible sulfide liquids with significantly different Ni abundances. The minimum number of magma pulses inferred from the number of sub-parallel olivine compositional arrays from a single drill core is two (see [Figs. 11, 12](#)).

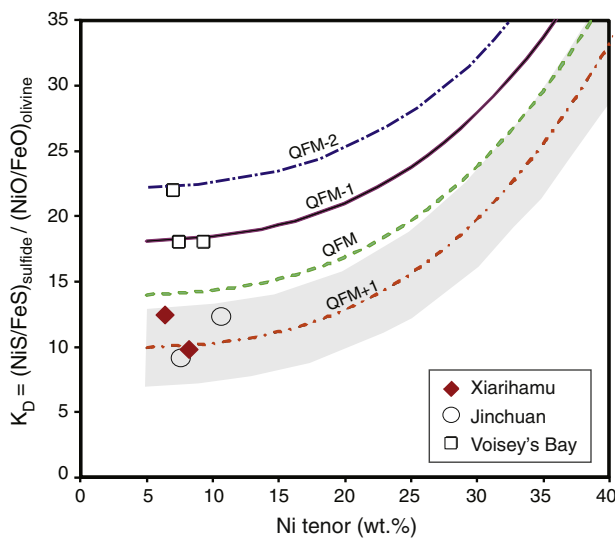


Fig. 17. The oxidation state of the Xiarihamu magmatic sulfide deposit determined using olivine compositions and Ni contents in bulk sulfide. The base of the plot and the results for Jinchuan and Voisey's Bay deposits are from [Barnes et al. \(2013\)](#).

Table 4

Estimated parental magma composition for the Xiarihamu lherzolites.

T, °C ⁽¹⁾	SiO ₂	TiO ₂	Al ₂ O ₃	Fe ₂ O ₃ ⁽²⁾	FeO	MnO	MgO	CaO	Na ₂ O	K ₂ O	P ₂ O ₅	Total
1270	52.42	1.28	15.60	1.75	6.54	0.64	9.79	8.91	1.88	0.90	0.30	100.00

⁽¹⁾The temperature at which olivine starts to appear on the liquidus under the conditions of 1 kb and FMQ + 1 oxidation state.⁽²⁾Calculated from total iron in the magma at 1 kb, 1270 °C and FMQ + 1 oxidation state using the MELTS program.

6.5. Conceptual genetic model and exploration implications

The results from this study show that the Xiarihamu ultramafic rocks are characterized by light REE enrichments and pronounced negative Nb anomalies in whole rocks, Ca-depletion in olivine, and high SiO₂ in the parental magma. These features, together with regional geological constraints and age data given above, support a model whereby the Xiarihamu ultramafic intrusion is the product of subduction-related boninitic magmatism at ~412 Ma in the Kunlun arc (Fig. 19a). We propose that the primitive arc magma became contaminated with crustal sulfur during magma ascent, resulting in sulfide saturation in the magma. The ascending magma was charged with small droplets of immiscible sulfide liquid. Upon arrival at the Xiarihamu dynamic conduit system, the immiscible sulfide liquid, together with some olivine crystals, settled down to form the sulfide–olivine cumulates while the magma continued to ascend to the higher levels (Fig. 19b). This process repeated several times to form multiple sulfide-bearing zones and alternating occurrence of sulfide-poor and sulfide-rich zones. Similar Fo contents but significantly different Ni contents in olivine from different sulfide zones (Figs. 12a, 14a) reflect different sulfide liquid compositions during the development of the system, i.e., the higher the Ni content in the sulfide liquid, the higher the Ni content in the olivine with the same Fo content after Fe–Ni exchange between these two phases (see a summary in Li et al., 2007). The spatial variations cannot be explained by in situ fractional segregation of immiscible sulfide liquid from magma because the inferred Ni tenors do not decrease upward (Figs. 11, 12, 14). Other processes such as upgrading by new surge of magma through the Xiarihamu conduit or variable R-factors (magma/sulfide mass ratios, Campbell and Naldrett, 1979) at depth are more plausible.

The conceptual genetic model bears some significant implications for global and regional exploration for magmatic Ni–Cu sulfide deposits. Globally, the Xiarihamu discovery is an encouraging example for the exploration of magmatic Ni–Cu sulfide deposits in arc settings. Regionally, other mafic–ultramafic intrusions in the Kunlun arc terrane should be evaluated for their mineralization potential. The most favorable criteria are similar ages, similar lithology and similar geochemical signatures to the Xiarihamu sulfide ore-bearing ultramafic intrusions. Once such an

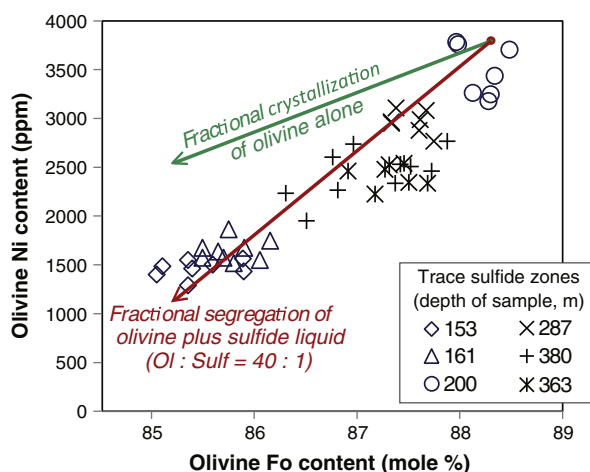


Fig. 18. Modeling of olivine fractional crystallization and concurrent sulfide segregation in the Xiarihamu magma. See text for explanation.

intrusion has been found, the exploration should focus on the areas where olivine is more abundant and where possible hydraulic traps in a feeder dike or sheet occur, such as at turning points and widen portions of the conduit (Li and Naldrett, 1999; Lightfoot and Evans-Lamswood, 2015).

7. Conclusions

New findings and important conclusions from this study are listed below.

- (1) Zircon U–Pb isotope age data show that the Xiarihamu ultramafic body is ~20 Ma younger than the associated gabbro intrusion.
- (2) The $\varepsilon_{\text{Hf}}(t)$ values of zircon crystals from the Xiarihamu ultramafic rocks are from 1 to 5, consistent with minor crustal contamination of the parental magma.
- (3) The Xiarihamu ultramafic rocks are characterized by light REE enrichments, pronounced negative Nb anomalies and Ca-depletion in olivine, which are common for arc cumulates.
- (4) The spatial variations of olivine Fo–Ni contents indicate that multiple pulses of sulfide-laden magma were involved in the development of the Xiarihamu magmatic sulfide system.
- (5) Elevated $\delta^{34}\text{S}$ values (3.5–6.8‰) for the deposit indicate that external sulfur played a critical role in sulfide mineralization.
- (6) The Xiarihamu sulfide ore-bearing ultramafic intrusion is the product of subduction-related, boninitic magmatism.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.lithos.2015.01.003>.

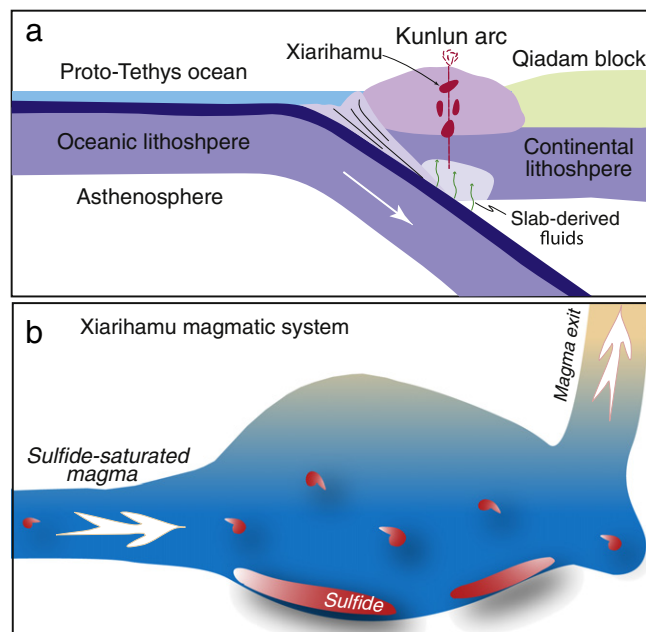


Fig. 19. Conceptual genetic models for the Xiarihamu magmatic sulfide deposit, (a) petrogenesis and (b) ore genesis. See text for explanation.

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